

object also contains an extra internal field called a *reference count*, which keeps track of how many different variables refer to that object. I show how this works in [Listing 11-1](#).

File: ReferenceCount.py

```
# Reference count example

❶ class Square():
    def __init__(self, width, color):
        self.width = width
        self.color = color

    # Instantiate an object
❷ oSquare1 = Square(5, 'red')
    print(oSquare1)
    # Reference count of the Square object is 1

    # Now set another variable to the same object
❸ oSquare2 = oSquare1
    print(oSquare2)
    # Reference count of the Square object is 2
```

Listing 11-1: A simple `Square` class for demonstrating reference counting

We can use Python Tutor (<http://pythontutor.com/>) to step through our code. We start with a simple `Square` class ❶ containing a few instance variables. We then instantiate an object and assign it to the variable `oSquare1` ❷. [Figure 11-1](#) shows what we see after instantiating the first object: as you can see, the variable `oSquare1` refers to an instance of the `Square` class.